NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY POLITEHNICA BUCHAREST

Predicting laptop prices in Romania from specifications and price dynamics

STUDENTS: LEVENT AMET
BOGDAN PETRIC

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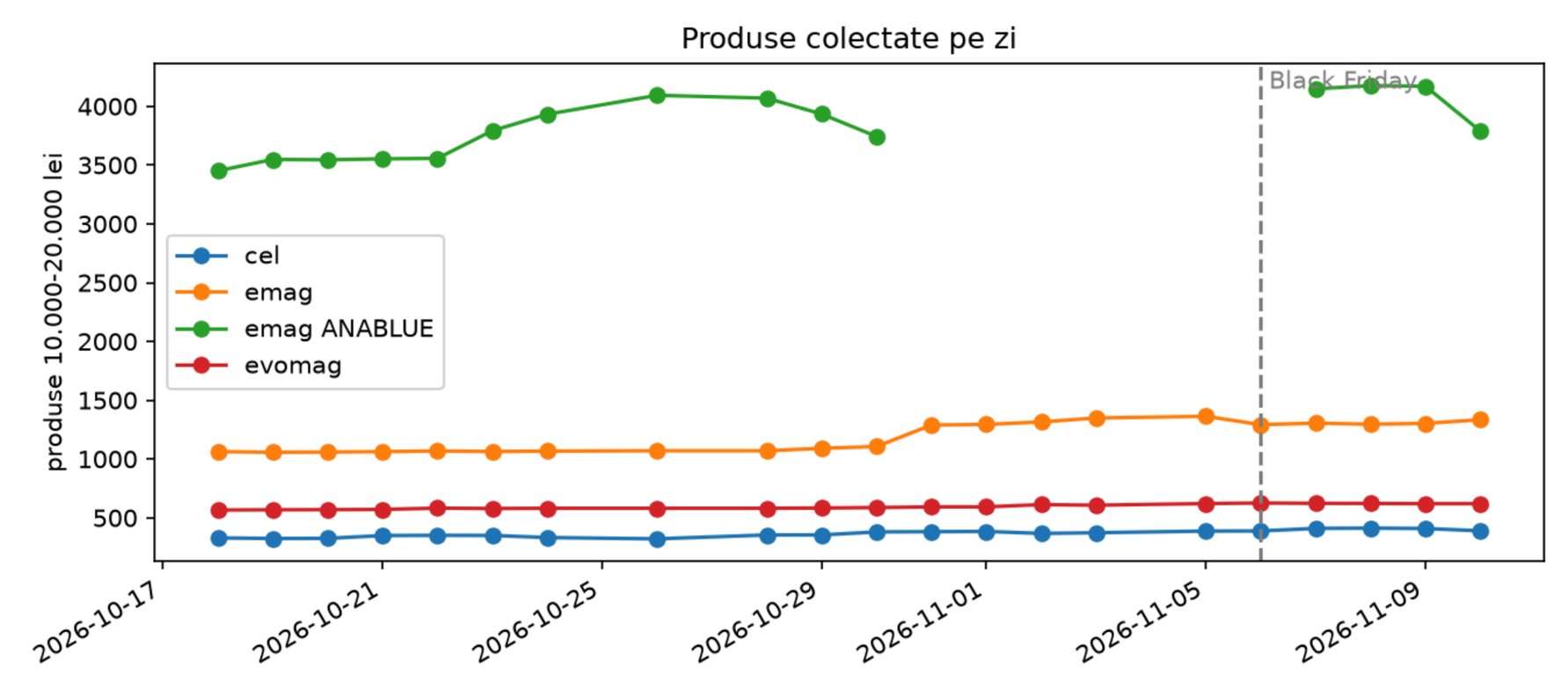
Problems & Goals

- **Problems:**
  - Laptop prices vary across shops and over time, especially around Black Friday and Christmas
  - Buyers can't easily tell whether a price is fair for the specifications
- **Goals:**
  - Price model. Estimate a laptop's price from its specs (CPU, RAM, storage, GPU, screen, brand)
  - When and where prices drop: Black Friday, discounts, differences between shops

Data acquisition

A daily crawler across 3 Romanian shops, since **18 Oct 2026**



Data structure

The crawler produces three types of files:

- **Daily prices:** One file per shop and day, one row per product, with the price, stock and seller. In total: 132,252 rows over 21 days.
- **Lookup tables:** part_numbers.csv links the same laptop across shops. black_friday.csv adds the Black Friday prices of sold-out deals.
- **products.csv:** Everything merged into one row per product: specifications, benchmark scores and a price summary.

All files are linked by shop and product ID.

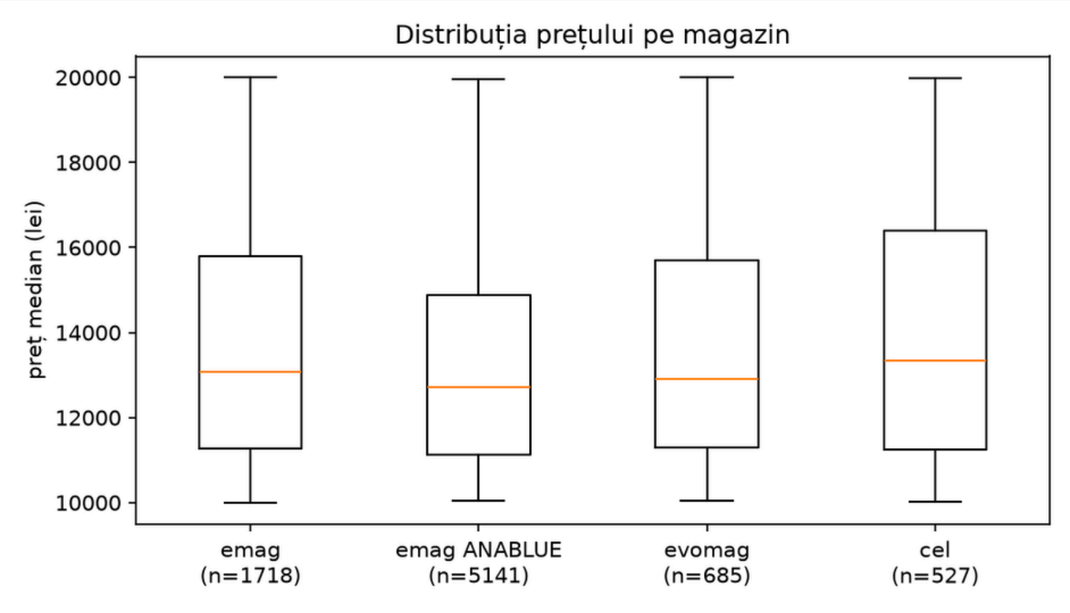
Crawling challenges

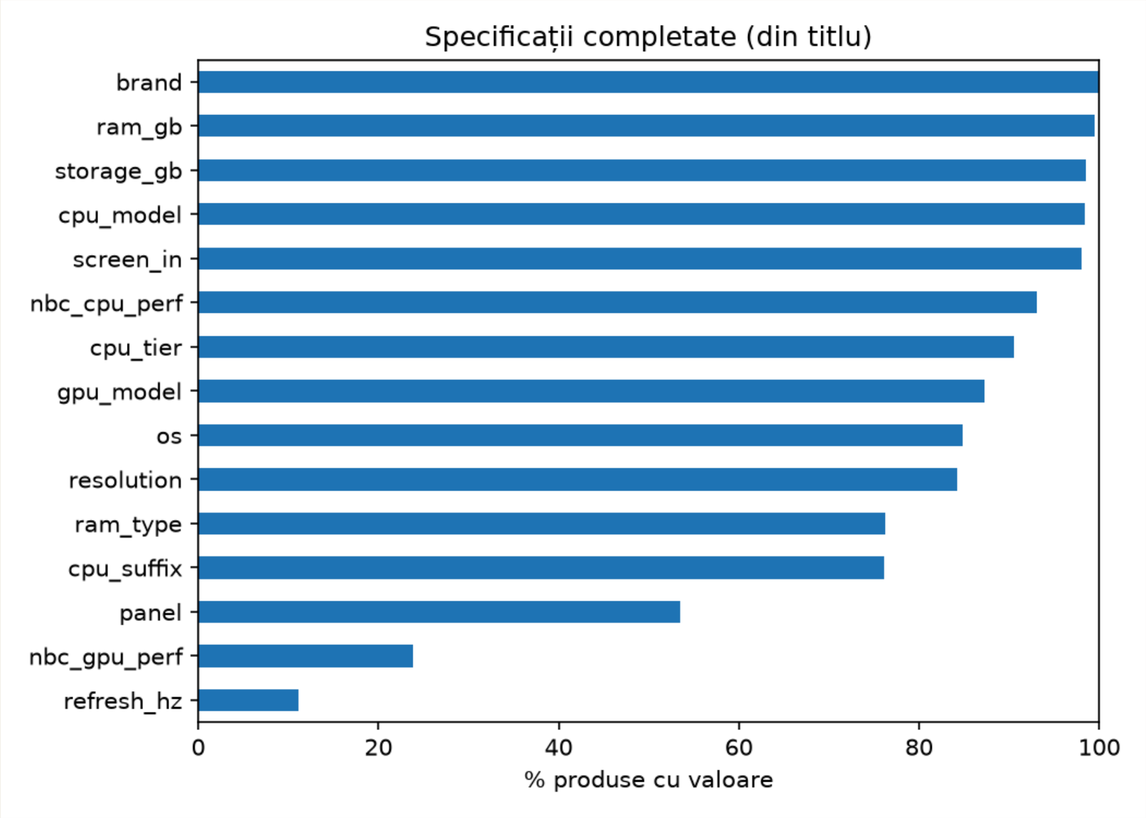
- **Same laptop has a different ID in every shop:** Manufacturer part number from the product page, normalized
- **Price filter dropped products exactly when discounted:** Collection range widened to 5,000–25,000 RON
- **One eMAG marketplace seller (ANABLUE) lists thousands of modified configurations (extra RAM/SSD, fake codes):** Detected by seller; analysed separately
- **eMAG blocked our IP many times (HTTP 511):** Daily page limit, longer delays
- **Black Friday deals sold out before our daily run:** Recovered the sold-out price from the product page banner

Dataset descriptives (1)

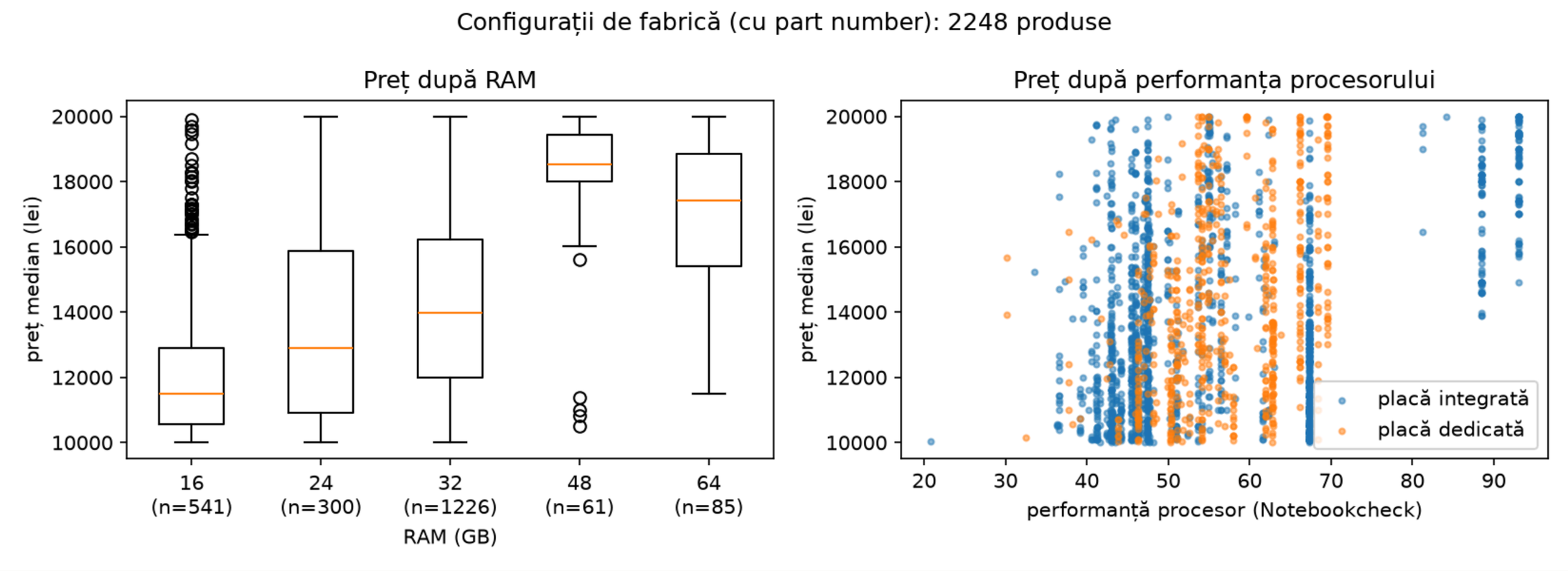
- 21 days collected (18 Oct – 10 Nov 2026)
- 132,252 daily price observations
- 14,137 unique products; 8,071 in the study range (median price 10,000–20,000 RON)
- 2,248 with a manufacturer part number; 510 found in 2+ shops
- 55 columns: 25 specs, 15 benchmark scores, price summary, Black Friday price

Dataset descriptives (2)

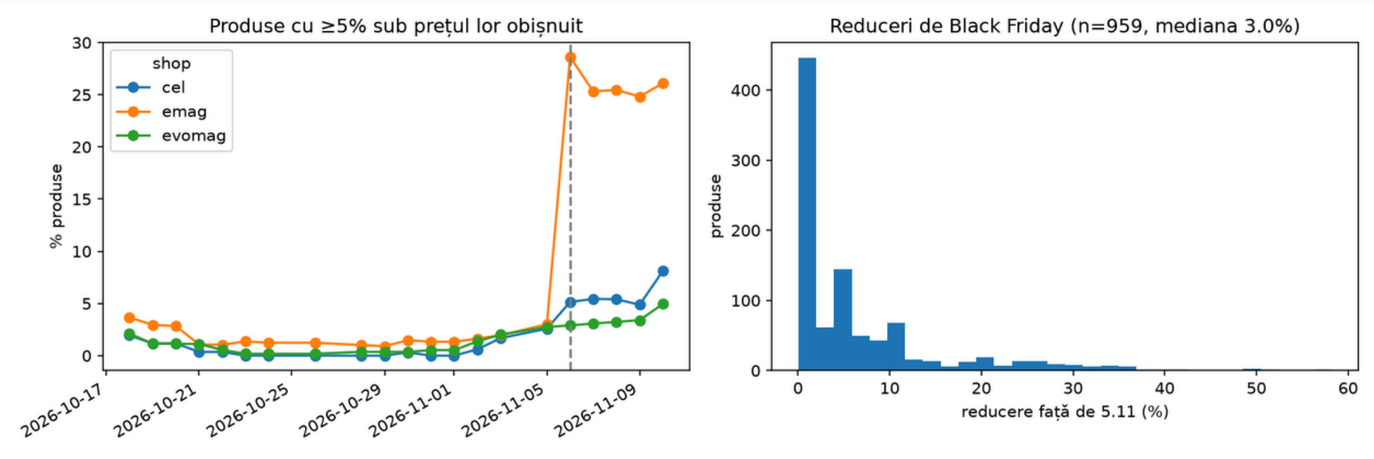




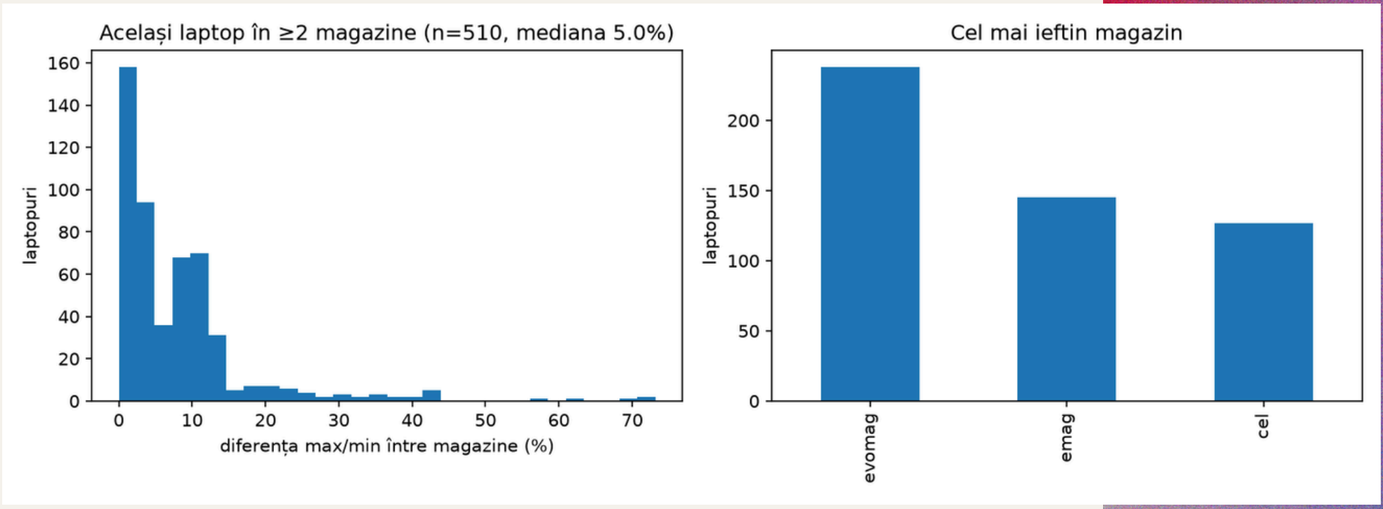
Dataset descriptives (3)



Price dynamics



- Black Friday discounts were concentrated at eMAG
- Most Black Friday discounts were small
- The same laptop can cost 5–10% more in another shop



Next steps: predicting price and the best time to buy

Price model:

- Predict a laptop's price from its specifications (brand, CPU, RAM, storage, GPU, screen).
- Start with a simple baseline, then Ridge regression, Random Forest and CatBoost.

Best time to buy:

- Check whether Black Friday discounts were real or prices were raised beforehand.
- Predict which laptops are likely to be discounted.
- Compare prices for the same laptop across shops.
- Keep collecting data until January, to use the Christmas sales as a second event.

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Thank you for your attention!

https://github.com/bogdanpetr/laptop-prices

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